

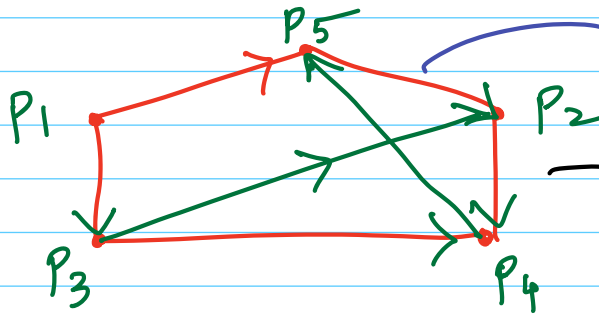
Lec. 09

Graph Theory + Social Networks

Adjacency matrix

Example: consider graph below:

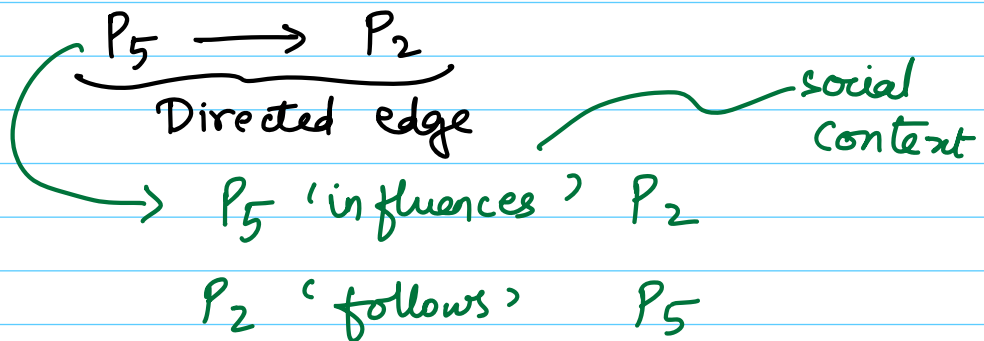
$$\begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 & 5 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{matrix} & \begin{bmatrix} 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 \end{bmatrix} \end{matrix}$$



This is a directed graph

This can represent a social network

$P_1, P_2, \dots, P_5$
people



'Adjacency matrix'

$$M = n \times n \quad \text{no. of nodes}$$

$$m_{ij} = 1 \quad \text{if there is a directed edge from } P_i \rightarrow P_j$$

Adjacency matrix: M

$$M^r = \underbrace{M \times M \times \dots \times M}_{r \text{ times}}$$

$[M^r]_{ij}$ ← no. of r step connections from P_i to P_j

no. of paths of length r from P_i to P_j .

For instance,

$$M^2 = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 & 5 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{matrix} & \begin{bmatrix} 0 & 2 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix} \end{matrix}$$

length $\downarrow$

$$[M^2]_{1,2} = 2 \rightarrow \text{no. of paths}$$

$\Rightarrow$ There are 2 paths of length 2 from P_1 to P_2

$$\begin{aligned} P_1 &\rightarrow P_5 \rightarrow P_2 \\ P_1 &\rightarrow P_3 \rightarrow P_2 \end{aligned}$$

$$M^3 = \begin{bmatrix} 0 & 0 & 0 & 2 & 1 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

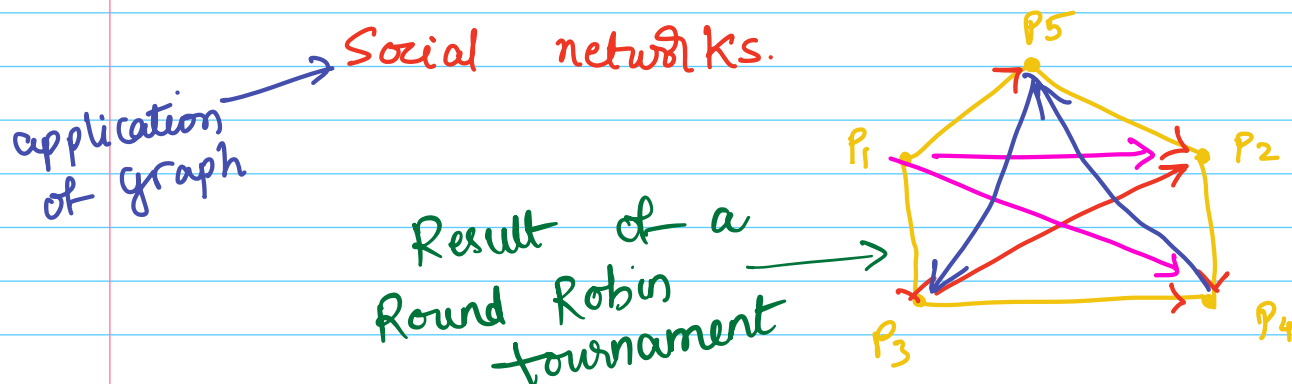
$$[M^3]_{1,4} = 2$$

$\Rightarrow$ 2 paths of length = 3 from P_1 to P_4

$$P_1 \rightarrow P_5 \rightarrow P_2 \rightarrow P_4$$

$$P_1 \rightarrow P_3 \rightarrow P_2 \rightarrow P_4$$

$[M^r]_{ij} \rightarrow$ # r step connections or
 r length path steps
 from P_i to P_j .



Between any pair of points P_i, P_j there
 is either edge from P_i to P_j or
 P_j to P_i , but NOT both.

$\rightarrow$ such a graph is known as a
 'Dominance Directed graph'.

$\rightarrow$ This is also known as a
 'Tournament'.

Dominance Directed Graph

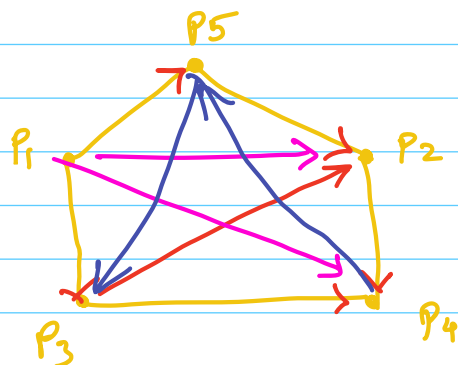
$\rightarrow$ which is the most influential node?

Consider $M + M^2$

adjacency matrix $\rightarrow$ Form sum of each row i

Vertex P_i with property row i has the largest sum
 = most influential node.

$$M = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 & 5 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{matrix} & \begin{bmatrix} 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 & 0 \end{bmatrix} \end{matrix}$$



$$M^2 = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 & 5 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{matrix} & \begin{bmatrix} 0 & 2 & 1 & 2 & 1 \\ \dots & \dots & \dots & \dots & \dots \end{bmatrix} \end{matrix}$$

$$M + M^2 = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 & 5 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{matrix} & \begin{bmatrix} 0 & 3 & 2 & 3 & 2 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 2 & 1 \\ 0 & 1 & 1 & 0 & 1 \\ 0 & 2 & 1 & 2 & 0 \end{bmatrix} \end{matrix}$$

10
 2
 4
 3
 5

row 1 has max. sum

$\therefore P_1$ is most influential person.